

	AMPACITY	MVA RATING	COND TEMP	SHEATH TEMP	TOTAL LOSS	STATUS
BURIED	323 A	18.46 MVA	90.0 °C	83.1 °C	21.276 W/m	Converged
FREE AIR	406 A	23.20 MVA	90.0 °C	79.1 °C	33.620 W/m	Converged

1. Design Data & Input Parameters

Parameter	Formula / Symbol	Value	Unit
Conductor material		Cu	
Conductor size		120	mm2
Conductor type		stranded	
Conductor diameter Dc		13.6	mm
Conductor screen OD Dc_sc		14.6	mm
Max conductor temp theta_max		90	degC
Insulation material		XLPE	
Insulation outer diameter Di		23.0	mm
Insulation screen OD Di_sc		24.1	mm
Insulation thickness t_ins		4.50	mm
Relative permittivity er		2.5000	
Loss tangent tan(delta)		0.001000	
Screen type		Cu_wire	
Screen mean diameter ds		27.3	mm
Screen area As		35.0	mm2
Armour type		none	
Oversheath material		PVC	
Cable outer diameter De		43.7	mm
Rated voltage U		33.0	kV
U0 line-to-earth		19.053	kV
Frequency		50	Hz
Bonding		solid	
Formation		trefoil	
Depth to cable centre L		1000	mm
Axial spacing s		43.0	mm
Ambient ground temp theta_a		30	degC
Soil thermal resistivity rho_soil		1.20	K.m/W
Ambient air temp theta_a_air		40	degC
Air mounting		free	
Solar radiation		None	
Wind speed		None	

2. DC Resistance [IEC 60287-1-1 Cl.2.1.1]

Parameter	Formula / Symbol	Value	Unit
IEC 60228 R0 at 20 degC		1.5300e-04	ohm/m
Temperature coefficient a20		3.9300e-03	1/K
R0 at theta_amb = 30 degC		1.5901e-04	ohm/m
R0 at theta_max = 90 degC		1.9509e-04	ohm/m

3. AC Resistance at theta_max [IEC 60287-1-1 Cl.2.1.2, 2.1.4]

Parameter	Formula / Symbol	Value	Unit
ks strand constant (IEC 60287-1-1 Table 2)		1.0000	
kp proximity constant (IEC 60287-1-1 Table 2)		0.8000	
$xs^2 = ks \cdot (8\pi f / R0) \cdot 1e-7$		0.6441309801	
$xs_{\text{■}} = (xs^2)^2$		0.4149047195	
ys skin effect factor		0.0021572327	
$xp_{\text{■}} = (xp^2)^2$		0.2655390205	
yp" intermediate proximity		0.0013814872	
yp proximity factor		0.0006051956	
R at theta_max AC = $R0(1+ys+yp)$		1.9562922296e-04	ohm/m

4. Dielectric Losses [IEC 60287-1-1 Cl.2.2]

Parameter	Formula / Symbol	Value	Unit
Relative permittivity er		2.5000	
Loss tangent tan(delta)		0.001000	
U0 line-to-earth		19.0530	kV
Di insulation outer diameter		23.0000	mm
Dc_sc conductor screen OD		14.6000	mm
Capacitance C = $er / (18 \ln(Di/Dc_sc)) \cdot 10^{-9}$		0.305604	nF/m
$\omega = 2 \pi f$		314.159265	rad/s
$Wd = \omega C U0^2 \tan(\delta)$		3.4852691566e-02	W/m
TB880 GP7		Applied	

5. Screen / Sheath Setup [IEC 60287-1-1 Cl.2.3]

Parameter	Formula / Symbol	Value	Unit
Screen material		Cu	
Screen resistivity rho_s		1.7241e-08	ohm.m
Screen area As		35.0000	mm2
Wire/sheath outer OD D_sc_outer		26.9000	mm
Lay factor LFs		1.056329	
Screen Rs0 at 20 degC (with LFs)		5.2035e-04	ohm/m

6. Armour Losses [IEC 60287-1-1 Cl.2.4]

Parameter	Formula / Symbol	Value	Unit
Armour type		None	
lam2		0.000000	

7. Thermal Resistances [IEC 60287-2-1]

Parameter	Formula / Symbol	Value	Unit
rho_ins insulation		3.50	K.m/W
rho_sc semiconducting screens		2.50	K.m/W
rho_tape bedding tape under wire screen		6.00	K.m/W
T1_cs conductor screen		0.0282307988	K.m/W
T1_i insulation		0.2531605113	K.m/W
T1_si insulation screen		0.0185883522	K.m/W
T1_uwbt bedding tape under wire screen		0.0388236256	K.m/W
T1_raw = sum of all layers		0.3388032879	K.m/W
T1 correction x1.00 (foil=False, cover=65.2%)		0.3388032879	K.m/W
T2 buried (screen to foil tape)		0.0000000000	K.m/W
T3 reference diameter D_start		27.3000	mm
T3_raw oversheath (single-layer)		0.1675990162	K.m/W
T3 buried x1.6 (touching trefoil factor)		0.2681584259	K.m/W
T3 air x1.0 (no touching factor in air)		0.1675990162	K.m/W
u = 2L/De (buried)		45.766590	
T4 buried (soil)		2.2269159435	K.m/W
T4 air (iterative — see air section)		see air section	

8. Electrical Parameters

Parameter	Formula / Symbol	Value	Unit
Inductance L		0.229492	uH/m
Reactance X = omega L		72.0971	uOhm/m
Pos-seq resistance R1		203.7388	uOhm/m
Pos-seq impedance Z1		216.1192	uOhm/m
Capacitance C		0.305604	nF/m
Charging current Ic		1.8292	mA/m

9. Sheath Losses at Convergence (Buried)

Parameter	Formula / Symbol	Value	Unit
theta_s at convergence		83.084118	degC
Rs at theta_s		6.4935e-04	ohm/m
X sheath reactance		7.3024e-05	ohm/m
lam1' circulating current		0.0414538613	
lam1" eddy current		0.0000000000	
lam1 = lam1' + lam1"		0.0414538613	
lam2 armour		0.000000	

10. Ampacity — Direct Buried [IEC 60287-1-1 Cl.1.4.1.1]

Parameter	Formula / Symbol	Value	Unit
Delta_theta = theta_max - theta_amb		60.00	K
Etop numerator		59.9071358393	
Ebot denominator		5.7462e-04	
I = sqrt(Etop/Ebot) exact		322.8848440303	A
I rounded TB880 GP2		323	A
MVA = sqrt(3) x U x I		18.4553	MVA
Iterations		5	
Convergence		YES	

11. Temperature Profile — Buried

Parameter	Formula / Symbol	Value	Unit
theta_c conductor target		90.0	degC
theta_c back-calculated		90.000000	degC
theta_s sheath surface		83.084118	degC
theta_j cable outer surface		77.378896	degC
theta_a ambient ground		30.0	degC

12. Loss Budget — Buried

Parameter	Formula / Symbol	Value	Unit
Conductor Wc = I ² x R		20.395251	W/m
Sheath Ws = lam1 x Wc		0.845462	W/m
Armour Wa = lam2 x Wc		0.000000	W/m
Dielectric Wd		3.485269e-02	W/m
Total		21.275565	W/m

13. Free Air Parameters [IEC 60287-2-1 Cl.2.2.1]

Parameter	Formula / Symbol	Value	Unit
Formation		trefoil	
Mounting		free	
Z constant (IEC Table 2)		0.9600	
E constant (IEC Table 2)		1.2500	
g exponent (IEC Table 2)		0.2000	
De* cable outer diameter		43.70	mm
De* in metres		0.043700	m
^g	De*	0.534683	
h = Z/(De*) ^g + E		3.0454567401	W/(m ² K ^{5/4})
Solar radiation		No	
Solar intensity H		N/A	W/m ²
Solar absorption sigma		N/A	
Ambient air temperature		40.0	degC
Delta_theta = theta_max - theta_a_air		50.00	K

14. Free Air T4 Calculation [IEC 60287-2-1 Cl.2.2.1]

Parameter	Formula / Symbol	Value	Unit
$x = (\theta_j - \theta_a)^{1/4}$ converged		2.4047122164	$K^{1/4}$
$D\theta_s = x^4 = \theta_j - \theta_a$		33.4389351263	K
$T_4 = 1/(\pi \cdot De \cdot h \cdot x)$		0.9946107430	K.m/W
Solar term $\sigma \cdot De \cdot H \cdot T_4$		0.0000000000	K
Etop numerator (incl solar)		49.9535897585	
Ebot denominator		3.0318e-04	

15. Sheath Losses at Convergence (Air)

Parameter	Formula / Symbol	Value	Unit
θ_s at convergence		79.073635	degC
R_s at θ_s		6.4115e-04	ohm/m
X sheath reactance		7.3024e-05	ohm/m
$\lambda_{1'}$ circulating current		0.0419706241	
$\lambda_{1''}$ eddy current		0.0000000000	
$\lambda_{1} = \lambda_{1'} + \lambda_{1''}$		0.0419706241	
λ_{2} armour		0.000000	

16. Ampacity — Free Air [IEC 60287-1-1 Cl.1.4.4.1]

Parameter	Formula / Symbol	Value	Unit
$I = \sqrt{E_{top}/E_{bot}}$ exact		405.9100637425	A
I rounded TB880 GP2		406	A
$MVA = \sqrt{3} \times U \times I$		23.2009	MVA
Iterations (outer)		5	
Convergence		YES	

17. Temperature Profile — Free Air

Parameter	Formula / Symbol	Value	Unit
θ_c conductor target		90.0	degC
θ_c back-calculated		90.000000	degC
θ_s sheath surface		79.073635	degC
θ_j cable outer surface		73.438935	degC
θ_a ambient air		40.0	degC

18. Loss Budget — Free Air

Parameter	Formula / Symbol	Value	Unit
Conductor $W_c = I^2 \times R$		32.232454	W/m
Sheath $W_s = \lambda_{1'} \times W_c$		1.352816	W/m
Armour $W_a = \lambda_{2'} \times W_c$		0.000000	W/m
Dielectric W_d		3.485269e-02	W/m
Solar heat $\sigma \cdot De \cdot H \cdot T_4$		0.000000	W/m
Total		33.620123	W/m

19. Electrical Parameters

Parameter	Formula / Symbol	Value	Unit
Inductance L		0.229492	uH/m
Reactance X = omega L		72.0971	uOhm/m
Pos-seq resistance R1		203.7388	uOhm/m
Pos-seq impedance Z1		216.1192	uOhm/m
Capacitance C		0.305604	nF/m
Charging current Ic		1.8292	mA/m



Standards: IEC 60287-1-1:2023 | IEC 60287-2-1:2015 | CIGRE TB 880 guidance applied.
GP2: ampacity rounded down. GP6: eddy losses always calculated. GP7: dielectric losses always calculated. GP8: exact T4 formula. T3: 1.6x factor for touching trefoil (buried only).
Solar radiation: not included.
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